Credit Risk Prediction Using Machine Learning: An Intelligent Agent Approach

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1. Introduction and Problem Definition

Credit risk assessment is a supervised decision-making problem in financial services where the objective is to classify loan applicants as either good credit risk (likely to repay) or bad credit risk (likely to default). This study frames the problem using the agent–environment–objective paradigm introduced in Coursework 1.

1.1 Agent

The agent is a machine learning classifier embedded within a lending decision pipeline. Two agents are implemented and compared: Logistic Regression and Decision Tree Classifier**.** Each agent receives an applicant's feature vector and outputs a binary prediction indicating credit approval risk.

The agent's rationality is defined as maximising expected utility, which in this context means correctly identifying high-risk applicants while minimising costly defaults.

1.2 Environment

The environment is the UCI Statlog (German Credit) dataset, consisting of 1,000 records with 24 encoded attributes describing financial history, employment status, credit behaviour, and personal information. The dataset is static and fully observable at decision time, making it a non-sequential supervised learning environment.

The dataset is moderately imbalanced (approximately 70% good credit, 30% bad credit), which introduces bias in naive accuracy-based evaluation and reflects real-world credit risk distributions.

Dataset Link - https://archive.ics.uci.edu/dataset/144/statlog+german+credit+data

1.3 Objective

The primary objective is binary classification performance, evaluated using F1-score, ROC-AUC, precision, and recall. However, rational decision-making is shaped by asymmetric misclassification costs, where false negatives (approving a bad applicant) are significantly more costly than false positives (rejecting a good applicant). Therefore, optimal performance requires balancing recall for bad credit against overall discrimination ability.

1.4 Constraints

- **Interpretability**: Decisions must be explainable to regulators and customers.
- **Computational efficiency**: Models must run on standard banking infrastructure.
- **Fairness**: Decisions must avoid discriminatory outcomes or proxy bias.
- **Cost sensitivity**: Misclassification costs are unequal and business-dependent.

2. Methods

2.1 Dataset and Preprocessing

The dataset was sourced in numeric format, containing 24 features and one binary target variable. The target was remapped from {1,2} to {0,1} for standard classification conventions.

Adding Feature Names to data

```
columns = [f"Feature_{i}" for i in range(1, 25)] + ["Target"]
df.columns = columns
```

```
df.head()
```

	Feature_1	Feature_2	Feature_3	Feature_4	Feature_5	Feature_6	Feature_7	Feature_8	Feature_9	Feature_10	...	Feature_16	Feature_17	Feature_18	Feature_19	Feature_
0	1	6	4	12	5	5	3	4	1	67	...	0	0	1	0	
1	2	48	2	60	1	3	2	2	1	22	...	0	0	1	0	
2	4	12	4	21	1	4	3	3	1	49	...	0	0	1	0	
3	1	42	2	79	1	4	3	4	2	45	...	0	0	0	0	
4	1	24	3	49	1	3	3	4	4	53	...	1	0	1	0	

5 rows × 25 columns

Figure 1: Dataset Overview after adding feature names

No missing values or duplicates were present. Exploratory analysis indicated moderate class imbalance and skewed feature distributions, motivating standardisation.

Check for Missing Values

```
df.isnull().sum()
```

	0
Feature_1	0
Feature_2	0
Feature_3	0
Feature_4	0
Feature_5	0
Feature_6	0
Feature_7	0
Feature_8	0
Feature_9	0
Feature_10	0
Feature_11	0
Feature_12	0
Feature_13	0

Figure 2: Applying Data Cleaning Steps

The dataset was split into 80% training and 20% testing using stratified sampling to preserve class distribution and reduce bias. This ensures representative evaluation on unseen data. Features were

standardised using StandardScaler fitted only on training data, preventing data leakage and ensuring valid, reliable model performance assessment.

Selecting Dependent and Independent Variables

```
X = df.drop("Target", axis=1)
y = df["Target"]
```

Train & Test Split

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

Figure 3: Train Test Split

This preprocessing is essential because distance-sensitive and gradient-based models such as Logistic Regression are sensitive to feature scaling.

2.2 Model 1: Logistic Regression

Logistic Regression is a supervised classification algorithm that estimates class probabilities using the sigmoid function. In this study, it was used as a baseline model due to its simplicity, interpretability, and computational efficiency. L2 regularisation was applied to improve generalisation, and hyperparameters were tuned using GridSearchCV with cross-validation. The parameter grid included different values of C and solvers such as lbfgs and liblinear, with a maximum iteration limit of 1000. The best performing configuration obtained was **C = 0.01, penalty = l2, and solver = liblinear**, evaluated using the F1 score to ensure balanced class performance.

```
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, f1_score, precision_score, recall_score, classification_report, confusion_matrix, roc_auc_score, roc_curve
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import GridSearchCV

# Define parameter grid
param_grid = {
    'C': [0.01, 0.1, 1, 10, 100],
    'penalty': ['l2'],
    'solver': ['lbfgs', 'liblinear']
}

# Logistic Regression model
lr = LogisticRegression(max_iter=1000)

# Grid Search for best parameters
grid_lr = GridSearchCV(lr, param_grid, cv=5, scoring='f1', n_jobs=-1)
grid_lr.fit(X_train, y_train)

# Best model
best_lr = grid_lr.best_estimator_

print("Best Parameters:", grid_lr.best_params_)
```

Figure 4: Logistic Regression Model

2.3 Model 2: Decision Tree Classifier

A Decision Tree is a supervised learning algorithm that recursively splits the feature space based on impurity reduction using criteria such as Gini index or entropy. In this study, it was selected due to its ability to capture non-linear relationships, provide interpretable decision rules, and operate without feature scaling. Hyperparameter tuning was performed using GridSearchCV with parameters including max depth, minimum samples split, minimum samples leaf, and criterion. The best configuration obtained was **criterion = entropy, max_depth = 15, min_samples_split = 5, and min_samples_leaf = 1**, optimised using F1 score with cross-validation for balanced performance evaluation.



```
# Parameter grid for tuning
param_grid = {
    'max_depth': [3, 5, 10, 15, None],
    'min_samples_split': [2, 5, 10, 20],
    'min_samples_leaf': [1, 2, 5, 10],
    'criterion': ['gini', 'entropy']
}

# Base model
dt = DecisionTreeClassifier(random_state=42)

# Grid Search with Cross Validation
grid_dt = GridSearchCV(
    dt,
    param_grid,
    cv=5,
    scoring='f1',
    n_jobs=-1
)

grid_dt.fit(X_train, y_train)

# Best model
best_dt = grid_dt.best_estimator_

print("Best Parameters:", grid_dt.best_params_)
```

Figure 5: Decision Tree Model

2.4 Evaluation Metrics

Given class imbalance and asymmetric misclassification costs, multiple metrics were used. Accuracy reflects overall correctness but is limited. Precision evaluates reliability of risk predictions, recall measures detection of defaulters, F1-score balances both, and ROC-AUC assesses overall discrimination across thresholds.

3. Results

3.1 Performance Summary

Logistic Regression outperforms the Decision Tree across all evaluation metrics, indicating stronger and more reliable predictive performance. The most significant difference is in ROC-AUC (0.801 vs 0.595), where Logistic Regression demonstrates strong discriminative ability, effectively separating good and bad credit applicants. In contrast, the Decision Tree performs only slightly better than random guessing, suggesting weak generalisation and limited usefulness in risk prediction.

Table 1: Summary of Results

Metric	Logistic Regression	Decision Tree
Accuracy	0.775	0.665
Precision	0.652	0.431
Recall	0.509	0.424
F1-Score	0.571	0.427
ROC-AUC	0.801	0.595

This trend is consistent across other metrics. Logistic Regression achieves higher accuracy (0.775 vs 0.665), precision (0.652 vs 0.431), recall (0.509 vs 0.424), and F1-score (0.571 vs 0.427). These results indicate that it not only makes more correct predictions overall but is also more reliable in identifying high-risk applicants. Overall, Logistic Regression aligns better with the objective of minimising financial risk in credit assessment.

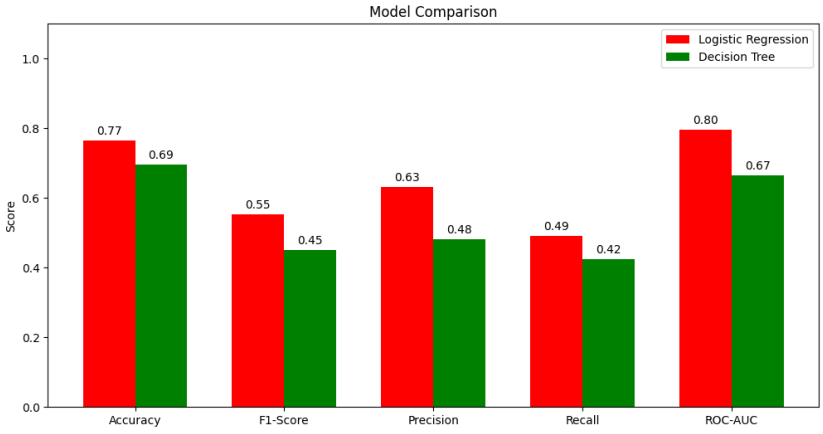


Figure 6: Model Results Comparison Plot

3.2 Confusion Matrix Analysis

The confusion matrix shows Logistic Regression correctly identifies most good applicants and some bad ones, but still produces false negatives, meaning risky applicants are approved, increasing potential financial losses.

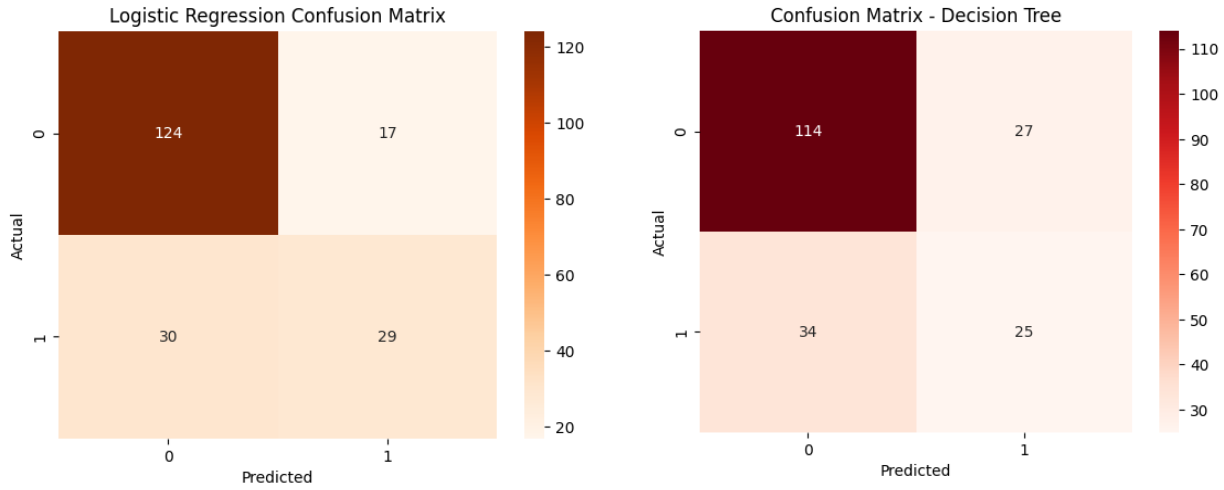


Figure 7: Confusion Matrix of models

In contrast, the Decision Tree produced both higher false negatives and false positives. This indicates weaker overall classification performance, as the model not only fails to identify risky applicants effectively but also incorrectly rejects creditworthy individuals. The imbalance between correct and incorrect predictions suggests poor generalisation.

3.3 ROC Curve Analysis

The Receiver Operating Characteristic (ROC) curve illustrates the trade-off between true positive rate and false positive rate across thresholds. Logistic Regression shows a strong curve above the diagonal with AUC 0.801, indicating good discrimination. In contrast, the Decision Tree's curve lies closer to the diagonal, with AUC 0.595, reflecting weak classification performance and poor generalisation on unseen data.

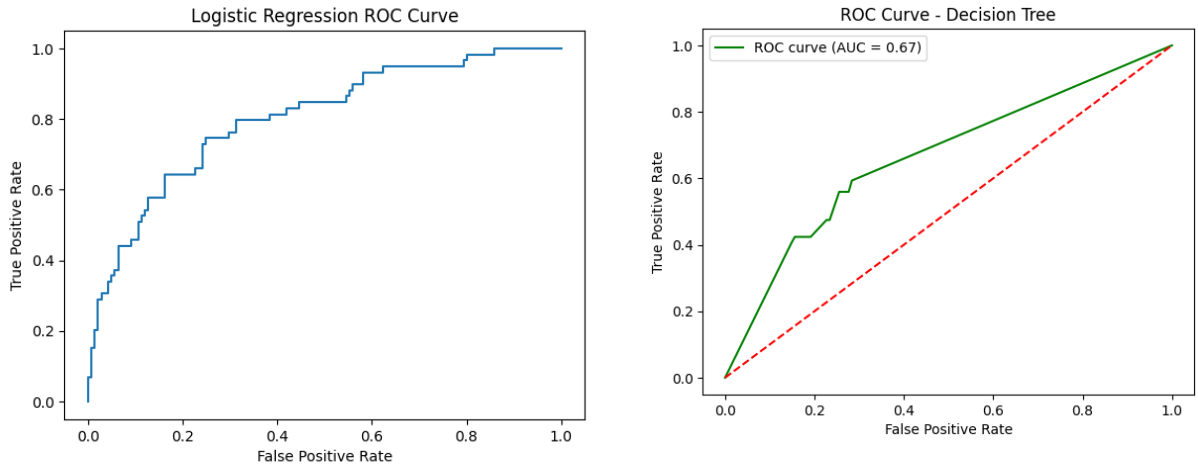


Figure 8: ROC Curves of models

3.4 Interpretation in Terms of Rational Behaviour

From an agent perspective, Logistic Regression behaves more rationally by maximising expected utility and effectively distinguishing risky from safe applicants. Its higher ROC-AUC and F1-score indicate better alignment with minimising defaults. The Decision Tree shows suboptimal rationality due to weak discrimination, unreliable decisions, and poor performance under uncertainty despite moderate accuracy.

4. Discussion

4.1 Rational Behaviour Analysis

Rationality here means maximising expected utility under asymmetric misclassification costs. Logistic Regression shows rational behaviour with strong discrimination (AUC ≈ 0.80), effectively separating credit risks, though moderate recall (~0.51) still allows some risky approvals. The Decision Tree is less rational, with near-random AUC (~0.59) and weak ranking ability.

4.2 Limitations

Logistic Regression assumes a linear decision boundary, limiting its ability to model complex feature interactions such as combined effects of income instability and loan size. Additionally, the default decision threshold (0.5) is not cost-optimised and may not reflect business risk preferences. The Decision Tree suffers from overfitting due to unconstrained depth, leading to poor generalisation on unseen data.

4.3 Alternative Approaches

More advanced but still interpretable methods include:

- **Random Forests**: Reduce variance through ensemble averaging and typically outperform single trees in credit scoring tasks.
- **Gradient Boosting**: Sequential learning improves weak decision rules and enhances predictive accuracy.
- **SVMs**: Provide strong decision boundaries but lack interpretability and require calibration for probabilistic outputs.

These methods were not implemented to maintain interpretability and alignment with coursework scope.

5. Ethics and Sustainability

5.1 Bias and Fairness

Credit scoring systems risk reinforcing historical inequalities present in training data. The German Credit dataset contains features such as age and personal status, which may act as proxies for protected attributes like gender. Even when protected attributes are excluded, correlated variables can still produce indirect discrimination. This raises concerns about proxy bias, where seemingly neutral features reproduce unequal outcomes across demographic groups.

False negatives (approving risky applicants) and false positives (rejecting creditworthy applicants) may disproportionately affect underrepresented groups due to imbalanced data representation.

5.2 Transparency and Regulation

Under modern regulatory frameworks such as the EU AI Act, credit decisions require explainability. Logistic Regression satisfies this requirement through interpretable coefficients, while Decision Trees provide rule-based transparency. However, explanations must be translated into human-understandable terms rather than raw feature indices.

5.3 Environmental Impact

Both models are computationally lightweight and require minimal energy for training and inference. Compared to deep learning approaches, classical models like Logistic Regression are significantly more sustainable and suitable for real-time financial deployment.

6. Future Work

6.1.1 Hyperparameter Optimisation

Hyperparameter optimisation using grid search can improve model generalisation. Tuning Logistic Regression regularisation strength and Decision Tree parameters such as max_depth, min_samples_leaf, and ccp_alpha helps control overfitting, enhance predictive accuracy, and produce more stable and reliable credit risk predictions.

6.1.2 Ensemble Methods

Ensemble methods such as Random Forests and Gradient Boosting can significantly enhance performance by combining multiple models. These approaches reduce variance, improve robustness, and capture complex non-linear relationships, leading to more accurate and generalisable credit risk predictions compared to single-model approaches.

7. Conclusion

This study evaluated two machine learning agents for credit risk classification using the German Credit dataset. Logistic Regression performed better, achieving strong discrimination (ROC-AUC ≈ 0.80) and stable generalisation, while the Decision Tree overfitted and showed weak performance. From an agent perspective, Logistic Regression shows partial rationality but still incurs misclassification costs. Overall, effective credit systems require balancing accuracy, interpretability, fairness, and cost sensitivity.

8. References

[1] S. Russell and P. Norvig, Artificial Intelligence: A Modern Approach, 4th ed. Hoboken, NJ: Pearson, 2021.

[2] H. Hofmann, "Statlog (German Credit Data)," UCI Machine Learning Repository, 1994. [Online]. Available: https://archive.ics.uci.edu/dataset/144/statlog+german+credit+data [Accessed: Apr. 2025].

[3] D. J. Hand and W. E. Henley, "Statistical classification methods in consumer credit scoring: a review," Journal of the Royal Statistical Society: Series A, vol. 160, no. 3, pp. 523–541, 1997. doi: 10.1111/j.1467-985X.1997.00078.x.

[4] A. Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 3rd ed. Sebastopol, CA: O'Reilly Media, 2022.

[5] C. M. Bishop, Pattern Recognition and Machine Learning. New York, NY: Springer, 2006.

[6] L. Breiman, J. Friedman, R. Olshen, and C. Stone, Classification and Regression Trees. New York, NY: Chapman and Hall/CRC, 1984.